5 Fig. 5.1 shows a ray of yellow light incident on a glass prism ABC.

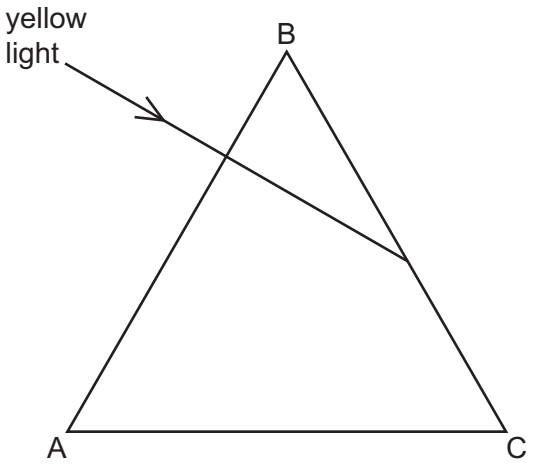

Fig. 5.1

(a) Explain why the ray does **not** change direction when it enters the prism at face AB.

.....

..... [1]

(b) The critical angle for the glass is 42°.

(i) Calculate the refractive index of the glass.

Show your working.

refractive index = ... [2]

(ii) On Fig. 5.1, continue the path of the light through the prism and after it leaves the prism. [3]

(c) Internet data can be transferred using infrared waves in optical fibres.

State **two** advantages of using optical fibres to transmit data.

1 .....

2 .....
[2]

[Total: 8]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN